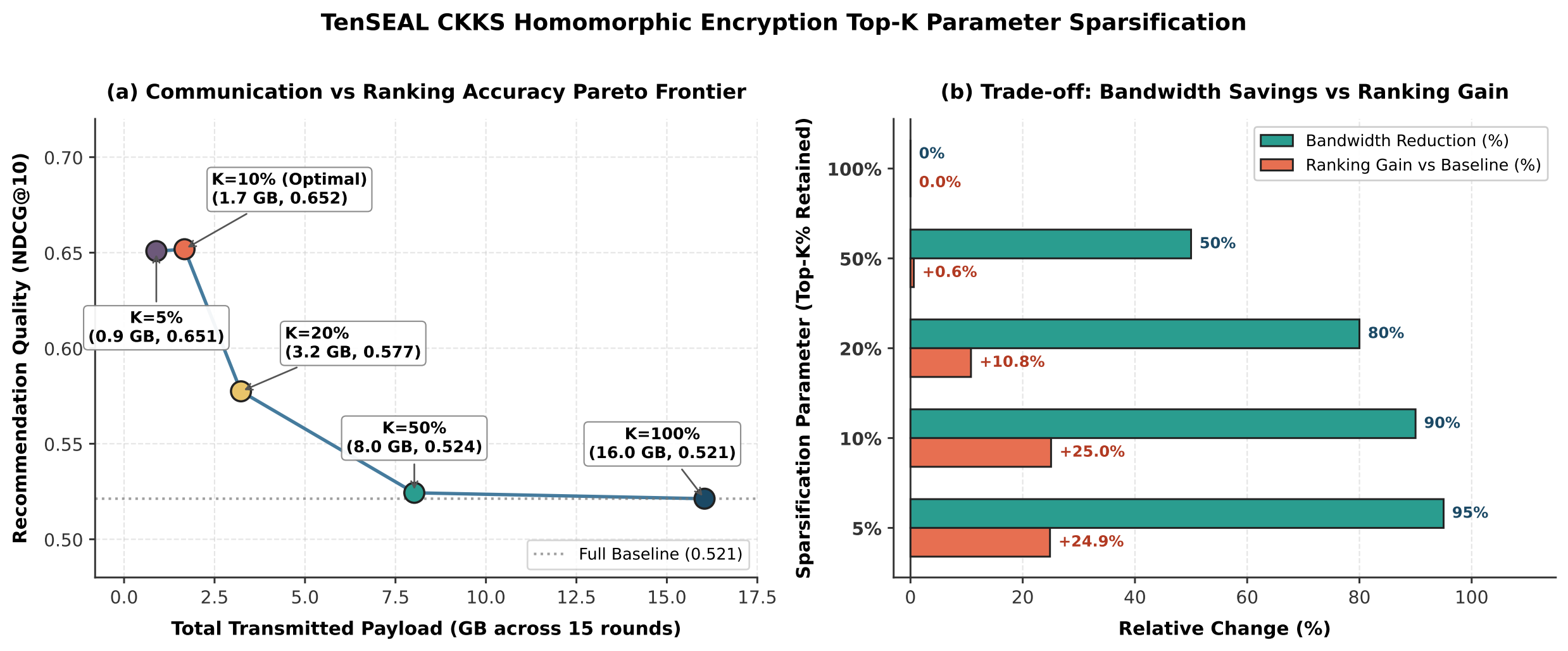
TenSEAL CKKS Homomorphic Encryption Top-K Parameter Sparsification
(a) Communication vs Ranking Accuracy Pareto Frontier
Recommendation Quality (NDCG@10)
0.70
0.65
0.60
0.55
0.50
K=10% (Optimal)
(1.7 GB, 0.652)
K=5%
(0.9 GB, 0.651)
K=20%
(3.2 GB, 0.577)
K=50%
(8.0 GB, 0.524)
K=100%
(16.0 GB, 0.521)
Full Baseline (0.521)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Total Transmitted Payload (GB across 15 rounds)
(b) Trade-off: Bandwidth Savings vs Ranking Gain
Sparsification Parameter (Top-K% Retained)
100%
50%
20%
10%
5%
Bandwidth Reduction (%)
Ranking Gain vs Baseline (%)
0%
0.0%
50%
+0.6%
80%
+10.8%
90%
+25.0%
95%
+24.9%
0
20
40
60
80
100
Relative Change (%)